

Models of interspecific interactions

Predator-prey dynamics and competition



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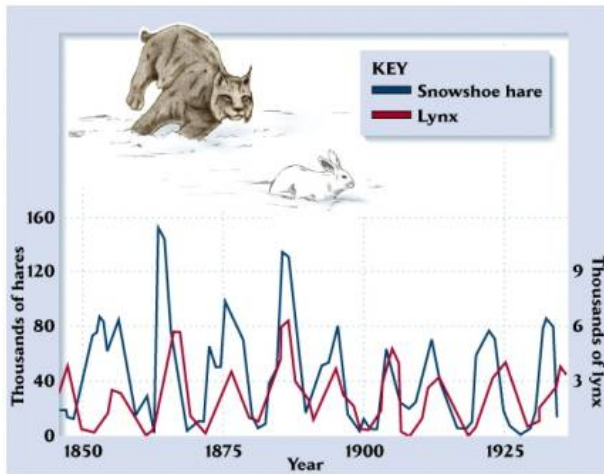
We will ignore potential extensions with stochasticity, age structure, spatial structure, etc. . .

How should predator-prey dynamics operate?

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LYNX-HARE CYCLES



Model for prey

$$N_{t+1}^{prey} = N_t^{prey} + N_t^{prey}(r^{prey} - k^{pred}N_t^{pred})$$

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- Model is based on geometric growth
- r^{prey} : growth rate of the prey in the absence of predators
- k^{pred} : kill rate
- b^{pred} : effect of prey on predator birth rate
- d^{pred} : predator mortality rate

Equilibrium for prey occurs when...

$$N^{pred} = \frac{r^{prey}}{k^{pred}}$$

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Equilibrium for predators occurs when...

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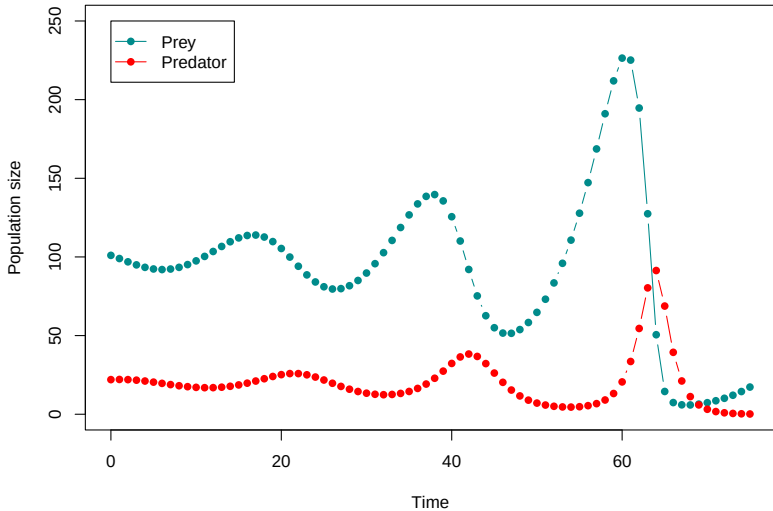
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Equilibrium for predators occurs when...

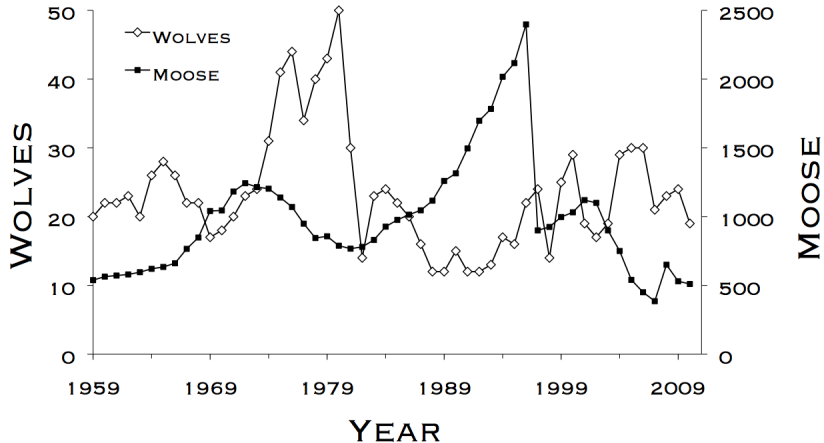
$$N^{prey} = \frac{d^{pred}}{b^{pred}}$$

However, it is rare that both equilibrium conditions will be met at the same time, and so the populations will cycle.

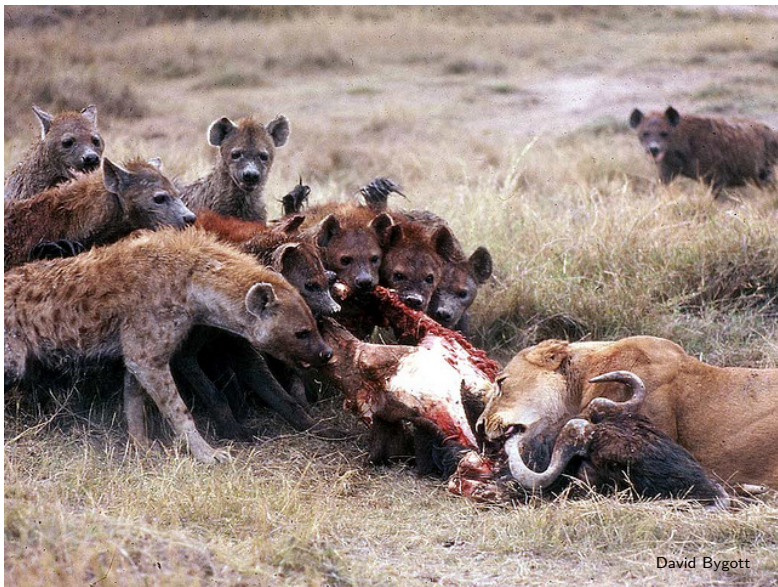
MODEL PREDICTS POPULATION CYCLES



ISLE ROYALE WOLVES AND MOOSE



<http://www.youtube.com/watch?v=PdwnfPurXcs>
<https://isleroyalewolf.org/>



Model for species A

$$N_{t+1}^A = N_t^A + r^A N_t^A (K^A - N_t^A - \alpha^B N_t^B) / K^A$$

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$$N_{t+1}^A = N_t^A + r^A N_t^A (K^A - N_t^A - \alpha^B N_t^B) / K^A$$

Model for species B

$$N_{t+1}^B = N_t^B + r^B N_t^B (K^B - N_t^B - \alpha^A N_t^A) / K^B$$

Model for species A

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Model for species B

$$N_{t+1}^B = N_t^B + r^B N_t^B (K^B - N_t^B - \alpha^A N_t^A) / K^B$$

- Model based on logistic growth
- The α parameters are competition coefficients determining how strongly each species affects the other

Equilibrium for species A

$$N^A = \frac{K^A - \alpha^B K^B}{1 - \alpha^A \alpha^B}$$

Equilibrium for species A

$$N^A = \frac{K^A - \alpha^B K^B}{1 - \alpha^A \alpha^B}$$

Equilibrium for species B

$$N^B = \frac{K^B - \alpha^A K^A}{1 - \alpha^A \alpha^B}$$

Outcomes depend on the sign of the numerators

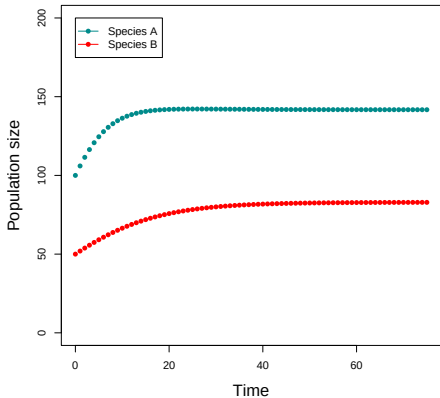
- (1) Stable coexistence
- (2) Competitive exclusion
- (3) Unstable equilibrium

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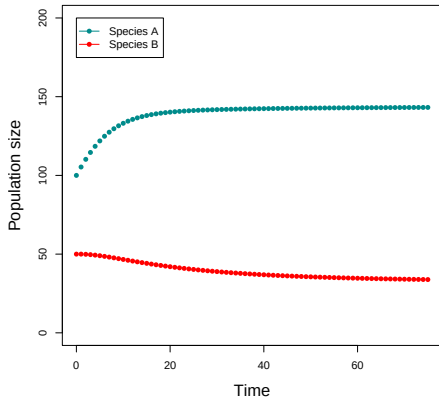
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Competitive exclusion principle: Two species with the same niche cannot coexist on the same limiting resource

Stable coexistence



Competitive exclusion



DON'T FORGET ABOUT INTRASPECIFIC COMPETITION



<https://youtu.be/KQLPL1qRhn8>

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These models could be extended to include:

- More species
- Stochasticity
- Age structure
- Harvest
- Spatial structure
- Additional forms of density dependence